🧠 Part 2: Case Study Analysis – AI Bias and Ethical Deployment *(40%)*

Case 1: Amazon's Biased Hiring Tool

🔍 Source of Bias

Amazon's AI recruiting system was trained on **10 years of historical résumé data**, primarily from male candidates in technical roles. The model absorbed **historical hiring biases**, penalizing resumes that included indicators associated with women (e.g., "women's colleges" or female-oriented organizations), thereby embedding **gender-based discrimination** from the past into future decision-making.

🛠️ Proposed Fixes

1. **Balanced & Representative Training Data**
   Curate a **diverse and inclusive dataset**, ensuring equal representation across **gender, race, educational background, and career paths**, to correct for historical overrepresentation and bias.

2. **Bias Auditing & Feature Neutralization**
   Regularly **audit outputs** for gender and demographic discrimination using **automated fairness tools**. Remove or mask features (like gendered language or institution names) that correlate with bias. Integrate **fairness constraints** into model design.

3. **Human-in-the-Loop Oversight**
   Maintain a **hybrid decision-making structure** where AI acts as a screening assistant, while trained human reviewers make the final hiring decisions, focusing on **inclusive hiring best practices** and ethical accountability.

Fairness Evaluation Metrics

- **Demographic Parity**: Ensures all groups have similar selection rates.
- **Equal Opportunity**: Confirms similar true positive rates for qualified candidates across groups.
- **Bias Impact Score**: Quantifies how bias changes after applying corrective strategies.
- **Disparate Impact Ratio**: Compares impact on different demographics (ideal = 1.0).
- **Candidate Satisfaction Feedback**: Surveys applicants for perceived fairness and transparency in the hiring process.

Case 2: Facial Recognition in Policing

Ethical Risks

- **Wrongful Arrests**: Misidentification rates are higher for **Black, Brown, and Asian individuals**, leading to **false accusations and legal consequences**.
- **Privacy Violations**: Facial recognition often functions without consent or oversight, enabling **mass surveillance** that disproportionately affects **marginalized communities**.
- **Discrimination Amplification**: AI tools used in biased systems (like policing) can **reinforce existing inequalities**, such as over-policing of certain neighborhoods.

- **Lack of Accountability**: Proprietary systems lack transparency, making it difficult for individuals to challenge misidentification.
- **Erosion of Civil Liberties**: Continuous surveillance can chill free speech and protest, impacting **democratic freedoms**.

Recommended Policies

1. **Mandatory Accuracy Audits by Demographics**
   Require **third-party performance evaluations** across **race, gender, and age**. Ban systems with **high error rates** for minority groups. Include **public reporting** of error rates and validation studies.
2. **Consent & Transparency Laws**
   Enforce laws requiring **explicit disclosure** when facial recognition is used. Provide **opt-out mechanisms** and ensure users know how their data is stored, used, or shared.
3. **Restricted Use Cases with Judicial Oversight**
   Limit usage to **specific, high-priority investigations** (e.g., identifying missing persons or terrorist threats) under **court-issued warrants**. **Ban real-time surveillance** in public spaces and prohibit use for **predictive policing or immigration enforcement**.

Additional Safeguards

- **Community Impact Assessments** before deployment in any jurisdiction.
- **Independent Oversight Boards** to audit practices and handle complaints.
- **Training for Law Enforcement** on model limitations, bias awareness, and ethical use.

Conclusion

Both cases highlight how AI can unintentionally reproduce and amplify systemic bias when built or deployed without ethical foresight. To ensure **fairness, safety, and trust**, organizations must combine **technical solutions (e.g., diverse data, fairness metrics)** with **governance frameworks** (e.g., policy regulation, human oversight, public transparency). Proactive design and responsible deployment are essential for AI systems to serve all members of society equitably.